

Phase 1 - Research Planner

Team Guide to the Autonomous Research System

Purpose: explain what Phase 1 does, why each field exists, and what it passes to Phase 2.

In one sentence: Phase 1 does not research the answer. It converts a user's topic into a validated, machine-executable research strategy that tells Phase 2 what evidence to search for, how strong that evidence must be, and when research is sufficient.

1. Where Phase 1 Fits

Input: a research topic, for example: “Is RAG actually effective at reducing hallucinations in LLMs?”

Output: a structured research plan saved as research_plan.json.

Boundary: Phase 1 plans the investigation. It does not search the web, collect sources, extract evidence, or produce the final conclusion.

2. Phase 1 Flow

Step	Component	Role
1	Research topic	User supplies the question to investigate.
2	Planner prompt	Defines the research-planning responsibilities and rules.
3	Structured schema	Forces the model to return the required JSON shape.
4	Gemini Planner	Makes planning decisions: questions, evidence needs, source standards, budgets, etc.
5	Validation layer	Checks logical consistency, required fields, dependencies, source counts, and contracts.
6	research_plan.json	Validated plan becomes the execution contract for Phase 2.

3. What the Planner Decides

Planner Field	Why It Exists	Example from Current Run
Research Domain	Classifies the topic so evidence standards can be adapted to the field.	AI/ML
Research Mode	Determines the style of investigation: exploratory, comparative, causal, evaluative, or descriptive.	EVALUATIVE
Temporal Sensitivity	Indicates how important recent evidence is.	HIGH

Working Hypothesis	Creates a neutral, testable proposition that research can support, weaken, or qualify.	RAG reduces factual hallucinations, but effectiveness depends on retrieval and context conditions.
Competing Hypotheses	Prevents confirmation bias by representing plausible alternative outcomes.	RAG may shift errors rather than remove them.
Falsification Criteria	Defines what evidence would materially weaken the working hypothesis.	Comparable studies showing no reduction or increased errors.
Evaluation Criteria	Turns vague concepts such as 'effective' into observable/measurable criteria.	Hallucination rate, faithfulness, effect of retrieval noise.

4. Source Policy and Evidence Standards

The Planner does not select actual papers. It defines which kinds of sources Phase 2 should prefer and what counts as evidence.

- Tier 1 - Evidence: peer-reviewed primary research, conference papers, strong primary research preprints.
- Tier 2 - Evidence: systematic reviews/surveys, official technical reports, official documentation.
- Tier 3 - Discovery: industry research, technical blogs, forum discussions.

Current global minimums: 12 total sources, 8 primary sources, 6 peer-reviewed sources, 8 quantitative sources, and 4 counter-evidence sources.

Important: these are planning targets, not proof that those sources already exist. Phase 2 may ultimately report insufficient evidence if the targets cannot be met within the search budget.

5. How the Topic Was Broken Into Research Questions

Question	Priority	What It Investigates	Why It Matters
Q1 - Performance	HIGH	Does RAG empirically reduce hallucination rates versus non-retrieval LLMs?	Establish the core effectiveness claim.
Q2 - Limitations	HIGH	How do retrieval failures, irrelevant context, and noise affect hallucinations?	Test whether retrieval quality changes the result.
Q3 - Comparison	MEDIUM	Does effectiveness vary by model architecture, scale, and pre-training	Identify model-dependent boundary conditions.

		distribution?	
Q4 - Counter-evidence	HIGH	Where does RAG introduce or worsen hallucinations versus closed-book generation?	Actively challenge the positive hypothesis.

6. What Every Research Question Contains

- **ID / Type / Priority:** Identifies the question and determines how important its coverage is.
- **Question:** The focused sub-question Phase 2 must investigate.
- **Evidence Needed:** The information required before the question can be answered.
- **Primary Queries:** Direct search queries for strong evidence.
- **Secondary Queries:** Alternative terminology and mechanisms to improve recall.
- **Counter-Evidence Queries:** Searches deliberately designed to find contradictory or qualifying evidence.
- **Preferred Source Types:** Source categories Phase 2 should prioritize.
- **Source Requirements:** Minimum evidence coverage for that question.
- **Quantitative Fields:** Exact numerical/context fields to extract when quantitative evidence is required.
- **Experimental Context:** Conditions needed to interpret empirical results correctly.
- **Boundary Conditions:** Conditions under which the answer may change.
- **Dependencies:** Questions that logically need earlier questions to be investigated first.

7. Example: Q1 as an Execution Contract

Question: What is the empirical impact of RAG on hallucination rates compared with standard non-retrieval LLMs across established benchmarks?

- **Evidence:** model, dataset, benchmark, hallucination rate, accuracy, retrieval configuration, metric
- **Minimum coverage:** 4 sources; 3 primary; 3 quantitative; 1 counter-evidence source
- **Quantitative extraction:** model, benchmark, metric, baseline score, experimental score, absolute and relative difference
- **Context:** model size, dataset, retriever, embedding model, top-k, evaluation metric

8. Evidence Sufficiency

Level	Meaning
HIGH	Multiple strong independent sources agree.
MEDIUM	Useful evidence exists, but coverage, independence, or quality is limited.
LOW	Evidence is weak, indirect, or sparse.
CONFLICTING	Comparable evidence supports materially

	different conclusions.
INSUFFICIENT	Evidence cannot justify a conclusion.

9. Stopping Criteria and Research Budget

Phase 1 also tells Phase 2 when research may stop. The current plan requires minimum source coverage, quantitative evidence where applicable, counter-evidence, coverage of all high-priority questions, support for major claims, and investigation of contradictions.

Budget limits: maximum 3 search rounds and maximum 20 total queries.

Critical distinction: budget exhaustion is not evidence sufficiency. If Phase 2 reaches the budget without enough evidence, the correct result can be 'insufficient evidence' rather than a forced conclusion.

10. Claim-Evidence Contract

The plan requires later research to preserve traceability:

Claim -> Evidence -> Source -> Source Location -> Relationship -> Confidence

This is intended to keep the final answer auditable and to separate what a source directly reports from interpretation and cross-source conclusions.

11. What Phase 1 Does NOT Do

- It does not search the web.
- It does not find or rank actual URLs.
- It does not claim that any evidence has already been found.
- It does not extract numbers or quotations from sources.
- It does not verify citations.
- It does not decide that RAG works or does not work.
- It does not execute Phase 2.

12. Why the Current Output Is a Successful Phase 1 Run

- ✓ The topic was classified as AI/ML and evaluative.
- ✓ A neutral working hypothesis and competing hypotheses were generated.
- ✓ The vague word 'effective' was operationalized into measurable evidence criteria.
- ✓ Four distinct research questions cover effectiveness, limitations, model dependence, and counter-evidence.
- ✓ Each question has search strategies, evidence requirements, quantitative/context fields, and boundary conditions.
- ✓ Source quality, diversity, evidence sufficiency, stopping criteria, and budgets are explicit.

- ✓ The validated plan was saved to research_plan.json.
- ✓ The run ended after planning; Phase 2 was explicitly not executed.

13. Handoff to Phase 2

Phase 1 asks: What must we investigate, what evidence counts, and when is it enough?

Phase 2 asks: Which real sources satisfy that plan, and what evidence can actually be extracted from them?

Therefore, research_plan.json is the execution contract between the Planner and the Phase 2 search/evidence pipeline.

Team Takeaway

Phase 1 is successful when it produces a rigorous, internally consistent research plan - not when it answers the user's question. The quality of Phase 2 depends on this plan because it defines the questions, evidence standards, counter-evidence strategy, budgets, and stopping rules before any search begins.